An OLG Model of Conflict and Production

A Study of Conflict, Corruption, and Foreign Aid

Abstract

I build a model simulating the interaction between competing groups in a single country, where each group can spend time on peaceful labor and conflict. Similarly to other papers I show that conflict increases with potential gains and decreases with opportunity costs in wages. By endogenizing both the benefit and wages with a CES production function, I find that when capital and labor are complements, there is a large possibility of poverty traps to arise. I also provide a brief discussion on why foreign aid should be applied very carefully in settings with potential for conflict.

1 Motivation

Foreign aid is an important policy tool that has garnered interest among economists. The notion of a 'big push' needed to shift poor economies out of poverty traps and into a better equilibrium is one of the central ideas in growth theory. However, the effects of applying foreign aid in practice have been quite limited, if not controversially negative. Qian (2015) suggests that the objectives of donor countries appear to be of greater importance than the needs of recipients when determining transfers. Moreover, the setting tends to be very heterogeneous, as aid might take on a vast variety of forms.

A major question is whether foreign aid might, in fact, harm receiving economies by creating chronic corruption or conflict. In particular, aid related to natural resources might harm the economy, as it most likely would not represent the interests of the country. In this case, the donor country might not be genuinely concerned with aiding the recipient through a 'big push,' but rather may only be interested in the natural resource itself. Moreover, it would clearly be willing to pay the lowest price possible to gain access to the mines, resource deposits, or oil fields. Usually, it appears to be cheaper to pay a group of corrupt politicians instead of compensating the whole country as an economy. Moreover, in some cases, making deals with local bandits could be easier, cheaper, or more appropriate than contacting federal authorities (see, for example, the setting in Sánchez De La Sierra (2020)).

My project describes a simple OLG-setup with potential conflict in order to study the relationship between conflict, aid, corruption, and perhaps natural resources. My model includes two competing groups that both can spend time on labor or conflict. Peaceful labor brings wages, while winning a conflict grants access to a corrupt government and all the benefits that come with it. The conflict is clearly destructive, as it both distorts labor from work, and extinguishes part of the available capital. Similar to many theoretical papers on conflict, I find that participation in war increases with the value of the benefit and decreases with labor wages (the alternative cost).

My main contribution, however, is making both the winning benefit and the wages endogenous, as described by market forces and a constant elasticity of substitution (CES) production function. I find that poverty traps occur only when there is a large complementarity (greater than Cobb-Douglas) between capital and labor in the production function. I argue that this effect arises from an endogenous interplay between wages and production, as under complementarity, wages grow faster than total production with increases in capital. I also provide a brief discussion on why it is important *how* to aid countries out of this poverty trap, and what donors should avoid in scenarios when capital inflow might be correlated with corruption.

My project relates primarily to the literature on conflict and civil wars. The baseline models of non-democracy and revolutions were formulated in Acemoglu and Robinson (2006), with several extensions described throughout the various chapters of the book. Dube and Vargas (2013) include natural resources and alternative costs in the analysis and test their model on data from Colombian oil and coffee price shocks. There is a growing branch focusing specifically on the interaction between foreign aid and conflict. First, in a controlled correlational analysis, De Ree and Nillesen (2009) find that aid appears to reduce the probability of conflict continuation, but not affect the probability of conflict onset. Two political science papers provide different explanations for that effect. Nielsen et al. (2011) suggests that it is the *reduction* in foreign aid that shifts powers in a country and leads to conflict, while Savun and Tirone (2011) suggest that it is the *increase* in democracy aid that leads to democracy and peace. I aim to reconcile these two observations, potentially by building a multi-period model in which donors can threaten recipients of stopping aid. For a review of the topic, I refer to Garfinkel and Skaperdas (2007), Blattman and Miguel (2010), and Findley (2018).

My paper is also closely connected to the literature on foreign aid. Here, the main reference is Qian (2015), who investigates the issue from a descriptive perspective, shows that the lowest-income countries receive less than their comparatively less poor counterparts, and argues that the objectives of donor countries are often more important than the needs of the recipients. Galiani et al. (2017) show that for countries at the threshold, a 1 percentage point increase in aid appears to raise GDP growth by approximately 0.35 percentage points. At the same time, Findley et al. (2017) proposes that aid might be controlled by elites, leading to smaller gains for the poor. Tavares (2003) and Mohamed et al. (2015) claim that aid reduces corruption. Moreover, the second paper proposes that aid from different bilateral sources competes with each other and sometimes has opposite effects. Interestingly, when estimating effects in the direction from corruption to aid, Alesina and Weder (2002) suggest that more corrupt governments appear to receive more aid. A more recent paper by Isaksson and Kotsadam (2018), however, suggests the opposite, as Chinese projects apparently increase corruption. In a review paper, Aidt et al. (2021) propose a unifying theory, modeling choices regarding foreign aid.

2 Theory

I construct an extension of the simple OLG general equilibrium model. More specifically, I introduce two opposing groups competing for power, choosing between labor and conflict. In Subsection 2.1 I describe the basic setup of the model, in Subsection 2.2 I solve it, and in Subsection 2.3 I discuss some policy implications from that model.

2.1 Setup

My model describes a polarized country with two similarly sized groups. Specifically, let there be a group A of size $\frac{1}{2}$ and a group B also of size $\frac{1}{2}$. Each individual in groups H and T has exactly one parent and exactly one child. Every individual lives for one period. Each period consists of two steps. In the first step, the groups choose the allocation of worker-time between fighting and working, and in the second step, individuals consume, make children, leave a bequest, and die.

Both groups can allocate their total worker-time of $\frac{1}{2}$ between labor supply l_i and conflict g_i. In the labor market, both groups earn a wage w determined by the firms' optimization decisions. In conflict, the investment of time affects the probability of winning a war, defined as $p_i = \frac{g_i}{g_i+g_{-i}}$. If the group wins the war, they pocket all the tax money.

The economy is endowed with k capital left as a bequest by the old generation, with part of it destroyed during conflict. The production function is of the CES type. Namely, $f(k,l) = A(\alpha k^\theta + (1-\alpha)l^\theta)^{\frac{1}{\theta}}$, where k is the total capital, $l = l_A + l_B$ is the total labor supply, and $\theta \leqslant 1$ is a parameter that can be interpreted as minus the elasticity of substitution. That means that the higher θ is, the more complementarity there is between capital and labor. A single perfectly competitive firm maximizes profit of $f(k,l) - wl$, where w is the wage. Clearly, in equilibrium, that wage is equal to $D_l f(k,l)$.

The government collects $V = tf(k,l)$ taxes, and all this money is spent on waste by the group in power. The two groups compete for this prize by investing effort in conflict. In expectation, each group receives $p_i V$ from participating in conflict. However, during conflict, part of the capital is destroyed, depending on the intensity of the conflict. Particularly, if $g = g_A + g_B$ of time is spent on war, then gk of capital gets destroyed.

Both groups maximize their expected utility as

$$\max_{l_i,g_i} wl_i + \frac{g_i}{g_i+g_{-i}}V, \quad s.t. \quad l_i + g_i = \frac{1}{2} \tag{1}$$

The equation of capital transition is given as

$$k' = (1-t)f[(1-g)k, l] \tag{2}$$

since t of output is spent on waste, while g of capital is destroyed.

2.2 Solution

Supposing an internal solution, the group maximization problem is easily solved as

$$w = \frac{g_{-i}V}{(g_i + g_{-i})^2} \implies g = \frac{V}{2w} \tag{3}$$

This is not my result, but a very common finding in the literature (Garfinkel and Skaperdas, 2007). Note that $l = l_A + l_B = 1 - g$ is spent on productive labor, and $(1-g)k$ of initial capital is left for production. Hence, we can easily find V and w as

$$V = tA\left(\alpha(1-g)^\theta k^\theta + (1-\alpha)(1-g)^\theta\right)^{\frac{1}{\theta}} = tA(1-g)(1-\alpha+\alpha k^\theta)^{\frac{1}{\theta}} \tag{4}$$

and

$$w = D_l f((1-g)k,l) = \frac{1}{\theta}A\left(\alpha(1-g)^\theta k^\theta + (1-\alpha)(1-g)^\theta\right)^{\frac{1}{\theta}-1}\theta(1-\alpha)(1-g)^{\theta-1} \tag{5}$$

or

$$w = (1-\alpha)A\left(1-\alpha+\alpha k^\theta\right)^{\frac{1}{\theta}-1} \tag{6}$$

By substituting 4 and 6 into 3, we get

$$g = \frac{tA(1-g)(1-\alpha+\alpha k^\theta)^{\frac{1}{\theta}}}{2(1-\alpha)A\left(1-\alpha+\alpha k^\theta\right)^{\frac{1}{\theta}-1}} = \frac{t\left(1-\alpha+\alpha k^\theta\right)}{2(1-\alpha)}(1-g) \tag{7}$$

or

$$g = \frac{\frac{t\left(1-\alpha+\alpha k^\theta\right)}{2(1-\alpha)}}{1+\frac{t\left(1-\alpha+\alpha k^\theta\right)}{2(1-\alpha)}}, \quad 1-g = \frac{1}{1+\frac{t\left(1-\alpha+\alpha k^\theta\right)}{2(1-\alpha)}} \tag{8}$$

which, fortunately, is always between zero and one, so we do not have to deal with corners. Now we can find a solved equation for the motion of capital as

$$k' = (1-t)f[(1-g)k, 1-g] = (1-t)(1-g)A\left(1-\alpha+\alpha k^\theta\right)^{\frac{1}{\theta}} \tag{9}$$

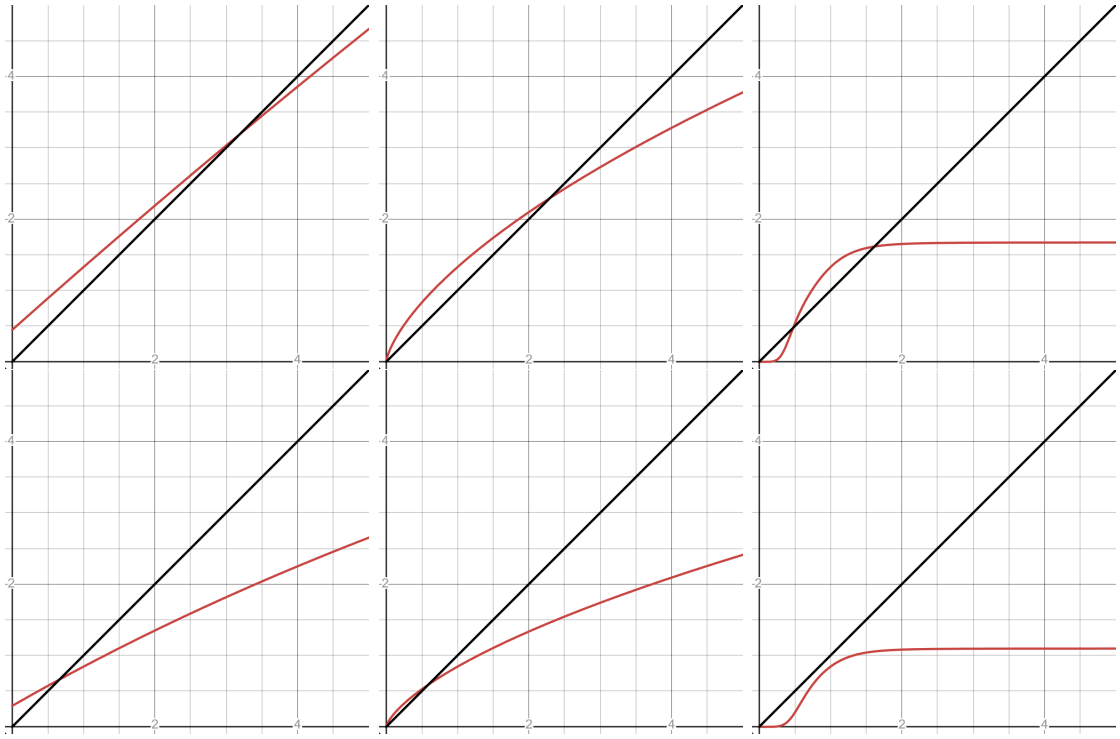
or

$$k' = A\frac{(1-t)\left(1-\alpha+\alpha k^\theta\right)^{\frac{1}{\theta}}}{1+\frac{t\left(1-\alpha+\alpha k^\theta\right)}{2(1-\alpha)}} \tag{10}$$

Now, while this equation is hard to analyze mathematically, it has a very interesting graphical representation that shows some nice intuition. In Figure 1, I present three graphs of capital motion with equal parameters of productivity A and tax waste t. However, what changes is the parameter of capital-labor complementarity.

By taking a very ugly second derivative from 10 and looking for the number of kinks in the function, I can claim the following statement:

Figure 1: Capital Transition for Substitution Params



This figure shows graphs of transition functions for $A = 1.5$, various tax waste rates of $t = 0.1$ (top) and $t = 0.4$ (bottom), as well as various substitution parameters of $\theta = 1$ (perfect substitutes, left); $\theta = 0$ (Cobb-Douglas, middle); and $\theta = -5$ (complements, right).

Proposition 1. *Under substitutability ($\theta > 0$), there is a single (stable) steady state. Under Cobb-Douglas ($\theta = 0$), there are two steady states (one stable, one at zero). Under complementarity ($\theta < 0$), and a large enough productivity parameter A, there are three steady states, two stable, meaning that one of them is a poverty trap.*

Note that this poverty trap arises without any fixed costs, simply through the groups having negative externalities on each other when fighting. Note also that complementarity is crucial for the emergence of multiple steady states. Intuitively, in this case, whenever capital is too small, wages are very low, and the poor have nothing else to do besides going on the streets to fight for the little they have left. However, with complementarity, wages start to grow faster than output (and war trophy V) with an increase in capital. Therefore, people start switching to labor work, less capital is destroyed, and there is rapid growth.

This complex interplay between conflict increase versus total production increase, as well as wage growth versus war trophy growth, allows for a convex part in the beginning of the transition function and a simultaneous concave part at the latter.

2.3 Discussion. Foreign Aid

We know that one usual solution to poverty traps is foreign aid – a 'big push' that shifts the country from bad equilibria to a good one, sometimes allocated through direct investment with subsidized interest rates for repayment. Importantly, sometimes this foreign aid is given directly to a government, making it more powerful and more attractive for a coup d'etat. In terms of my model, this would mean that the level of capital rises, but this, at the same time, is accompanied by larger 'tax rates' or simply a larger prize to win when gaining power. In particular, one could think of foreign investment in oil fields, which are very easy to tax and which give additional benefits to being in the government.

Note that by providing such institution-affecting aid, the donor does not only shift the economy from one equilibrium to another, but also changes the whole structure of the economy itself, which clearly also has long-run effects. Suppose, for example, that the economy is in a bad equilibrium of the top-right picture in Figure 1. Suppose also that a donor gives enough money for the economy to switch to the good state in the long run, provided that no other parameters change. Finally, suppose that, contrary to the donors' expectations, the parameters do indeed change, and that the government starts extracting more rent than before (say, corruption increases). In this case, instead of aiding the country into a better state, the donor aids the country into a harsh conflict, and in the end the economy even drops to zero instead of increasing to the good state or at least staying in the bad one.

Now let me consider a different scenario. Suppose that instead of aiding a poor country with capital, it were possible to aid this country with technology A. In this case, the production function would shift upwards, and the poor country might one day find itself in rapid growth towards a new steady state, without even having to invest anything into capital. This might appear a more suitable way to for future aid.

Note, however, that the interplay between foreign investment into k, tax rate t, and productivity A, is not modeled in any fashion formally in my project, and that most of my discussion in this Subsection is based on economic logic rather than on formal math. I am pretty sure, however, that it is quite possible to formalize my thought in later projects or other work.

3 Conclusion

To sum up, I have shown how complementarity between capital and labor can lead to the emergence of poverty traps in a setting with potential for civil conflict. I have also discussed some thoughts regarding foreign aid in similar settings, and argued that investment into technology and human capital (e.g schools) might be more attractive than simple monetary aid to the government, or even investment into capital. Most of this discussion, however, is based on some basic ideas arising from the model rather than direct formal derivation, which might have good potential for further work.

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